Quantitative Analysis of Media Bias and Stock Price Dynamics: The 2020 Shock

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***Abstract*—This paper studies how media reporting about large-cap firms changed around the 2020 recession and how firm-level media stance and stock returns relate to one another. We do not assume the influence runs in a single direction: we test both whether stance moves subsequent returns and whether returns move subsequent coverage. We compute daily target-dependent stance with a NewsMTSC pipeline from 690,586 scored news headlines (2015–2025) across 26 S&P 500 firms, keep the 90,579 that report a material firm event, and align this stance with firm-day returns and market controls. We ask whether firm-level media stance shifted after January 2020 (RQ1), whether firm-specific returns shifted (RQ2), and whether past stance and past returns help predict each other (RQ3). RQ1 and RQ2 use two-way fixed-effects pre/post panel regressions; RQ3 uses a weekly vector autoregression with Granger-causality tests. We find no significant aggregate shift in either stance or returns once common daily and monthly shocks are absorbed, and we find that the stance–return link is firm-specific rather than a market-wide regularity.**

I. Introduction

The 2020 recession is a natural breakpoint for studying the interaction between media narratives and financial markets. COVID-era financial news coverage expanded sharply in both volume and tone, and a growing body of evidence documents a structural change in news–market dynamics during and after the shock [4], [6]. This paper asks a single broad question: did the tone of firm-specific media coverage shift after 2020, and did that tone have a measurable relationship with firm-level stock returns?

This question matters because news is one of the main channels through which information about a firm reaches investors. If coverage systematically turns more negative or more positive, and if that tone carries information that prices do not already reflect, media tone should help explain firm returns; if prices already incorporate the same information, tone should add nothing once market factors are controlled. Separating these two cases is an empirical question about the strength and the direction of the media–market link, and we study it at the level of individual large-cap firms.

We measure tone with target-dependent stance rather than generic sentiment. A single news headline often names several firms, and a document-level sentiment score cannot separate positive coverage of one firm from negative coverage of another. Target-dependent stance instead scores the tone expressed toward a specific firm, which lets us build a daily, firm-level stance signal from news headlines and track how it behaves around the 2020 shock. We develop the stance model and the rest of the pipeline in the methodology.

Our empirical strategy has two parts. We first test whether firm-level stance and returns each shifted around January 2020 using pre/post panel regressions (RQ1 and RQ2), and we then test whether past stance and past returns help predict each other using a weekly vector autoregression with Granger-causality tests (RQ3).

The remainder of the paper proceeds as follows. Section II describes the data pipeline. Section III develops the methodology, including the formal hypotheses and estimating equations. Section IV presents the regression results for RQ1 and RQ2. Section V presents the VAR and Granger-causality analysis for RQ3. Section VI addresses identification and robustness, and Section VII concludes.

II. Data Construction

A. Corpus and Coverage

We construct a corpus of S&P 500 news coverage spanning 2015 to 2025. The records are collected through the MediaCloud news aggregation service: for each firm we query MediaCloud by company name, and the service returns the matching news items together with their publication, timestamp, headline, and article URL. The URLs let us recover additional page context where needed, but all downstream classification and scoring operate on the headline alone. The corpus contains 4,706,589 firm-headline records (Table I), and coverage is highly skewed across firms: a small number of heavily reported firms dominate the corpus, while many firms are covered only sparsely.

B. Data Preprocessing

We reduce the raw corpus in three steps. First, we discard headlines that do not name the firm, because matching a keyword against body text alone returns many articles that mention the firm only in passing; requiring the company name in the headline removes most of this contamination. Second, we remove duplicate headlines, because the same wire story is republished across many outlets. Third, we score the tone of each remaining headline toward the firm with a target-dependent stance model. These steps leave 690,586 scored firm-headline observations across 26 firms. A supervised relevance classifier, described in Section III-A, then keeps the 90,579 headlines that report a material corporate event, since only those headlines carry firm-specific stance. Averaging the stance of these relevant headlines for each firm on each day gives the daily stance panel of 27,627 firm-day observations used for RQ1. For RQ2 we merge each firm's daily stock return with market controls, which gives 62,667 firm-day return observations. Table II reports the count at each step.

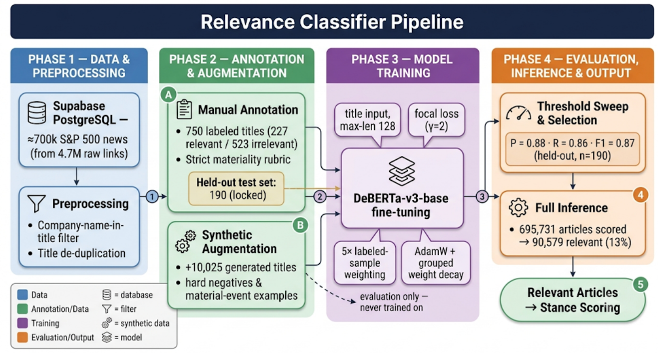

Fig. 1. Relevance classifier pipeline: raw S&P 500 news titles, preprocessing, manual annotation and synthetic augmentation, DeBERTa-v3 fine-tuning, threshold selection on a held-out test set, and full-corpus inference.

III. Methodology

A. Relevance Filtering and Annotation

A company name in a headline does not guarantee that the headline reports on that firm as a company. "Apple" can refer to a product, and many lifestyle or human-interest headlines name a firm without describing any corporate development. We therefore keep only headlines in which the firm is the primary subject of a material corporate event, such as earnings, mergers and acquisitions, executive changes, large layoffs, regulatory or legal action, major product launches, or credit and bankruptcy events. For example, "Goldman Sachs beats profit estimates on a trading surge" is relevant, whereas "Ten apple recipes to try this autumn" is not, even though both contain a firm name.

The filter has two stages: a title-level pre-filter that requires the spelled-out company name to appear in the headline, followed by a supervised classifier that judges materiality (Figure 1). To train the classifier we hand-labeled 750 headlines under this rubric, of which 227 are relevant and 523 are irrelevant. To keep the judgments consistent we labeled against the written rubric above, resolved borderline headlines by review, and locked a separate set of 190 labeled headlines as a test set that is never used for training or threshold selection.

The classifier is a DeBERTa-v3-base model fine-tuned for binary classification on the headline text. Each headline is tokenized to at most 128 tokens and padded to the batch length, and the model produces a 768-dimensional contextual representation that a linear head maps to a relevance probability. We train with the AdamW optimizer at a learning rate of 2×10^{-5}, a batch size of 16, weight decay of 0.01, and early stopping on the validation loss. Trained on the 750 labeled headlines alone, the model reaches a precision of 0.65 at a recall of 0.71. The main difficulty at this stage is the small label budget together with the gap between the labeled relevant rate, near 30%, and the true rate in the corpus, which is closer to 10% to 15%, so the model tends to over-predict relevance.

Because additional manual labels are costly, we expand the training set with 10,025 synthetic headlines generated programmatically rather than by a language model. A template generator fills sector-aware slots for each firm, so the events, amounts, and counterparties in a generated headline are plausible for that firm's industry. The synthetic headlines deliberately target the categories the base model confuses most: on the irrelevant side, consumer promotions, analyst previews, immaterial lawsuits, brand comparisons, and passing mentions; on the relevant side, earnings, layoffs, and deals phrased in unusual ways. We check their quality by sampling generated headlines against the rubric and by verifying that none duplicate the held-out test set. We then train on the combined set with focal loss, which concentrates learning on the hard boundary cases, and we weight the manual labels five times the synthetic ones, so that the synthetic data extends the decision boundary without overriding the human labels. This raises precision from 0.65 to 0.88 while holding recall above 0.85.

We adopt this single fine-tuned model as the final relevance classifier. On the held-out test set it attains a precision of 0.88, a recall of 0.86, and an F1 of 0.87 at its operating threshold, and a stricter threshold reaches a precision of 0.92 at a recall of 0.81 (Figure 2). Applied to the full corpus, the classifier labels 90,579 headlines as relevant, about 13%, which is consistent with the expected rate of material firm news. These relevant headlines are the input to the stance model described next.

Having defined the relevance filter, we can describe the coverage of the resulting analysis sample. Figure 3 shows the number of scored headlines per firm. Coverage spans more than four orders of magnitude, from AT&T with 139,701 scored headlines down to Eaton with 5, so a small subset of firms dominates the sample. The firm fixed effects in the panel regressions absorb these permanent differences in coverage volume.

B. Stance Scoring

We compute the tone of each relevant headline toward its firm with NewsMTSC, a target-dependent sentiment model. Given a headline and a target firm, the model returns a probability distribution over three classes, positive, negative,

TABLE I
YEARLY COUNT OF ARTICLES IN THE RAW CORPUS (2015–2025); THESE SUM TO THE 4,706,589 STARTING LINKS.

Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Total articles	570,999	625,582	652,741	525,259	559,027	363,962	291,144	260,115	204,281	204,007	449,472

TABLE II
SAMPLE CONSTRUCTION, FROM THE RAW CORPUS TO THE FIRM-DAY ESTIMATION PANELS.

Stage	Count
Raw firm-headline records (2015–2025)	4,706,589
After title filter, de-duplication, and stance scoring	690,586
Relevant (material-event) headlines	90,579
Distinct firms	26
RQ1 stance panel (firm-day relevant-stance observations)	27,627
RQ2 returns panel (firm-day returns)	62,667

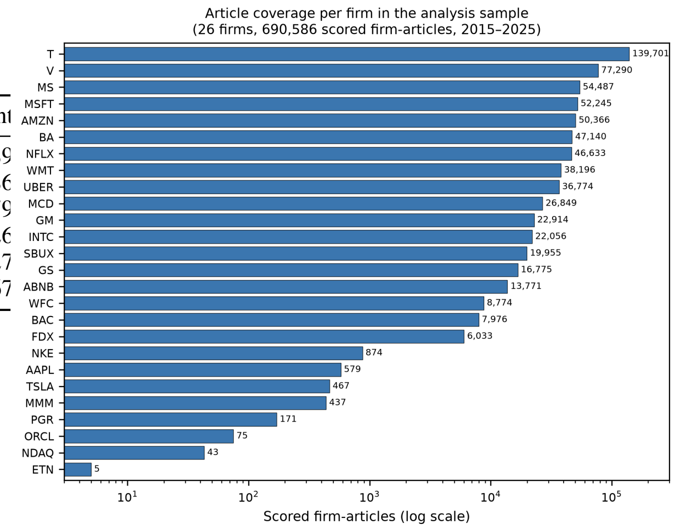

Fig. 3. Scored firm-articles per firm in the scored corpus (26 firms, 690,586 scored firm-articles from `articles_no_title_deduped`, 2015–2025; log scale).

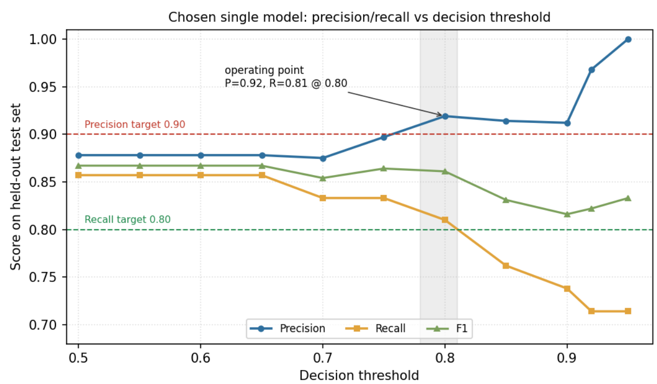

Fig. 2. Precision, recall, and F1 of the chosen single model on the held-out test set as a function of the decision threshold. A threshold near 0.80 satisfies both operating targets, reaching precision 0.92 at recall 0.81, the operating point adopted for full-corpus inference.

and neutral, describing the sentiment expressed toward that firm specifically rather than toward the headline as a whole. We write these class probabilities as p_a^+, p_a^-, and p_a^0 for headline a, with $p_a^+ + p_a^- + p_a^0 = 1$.

We convert each headline into a single signed stance score, $p_a^+ - p_a^-$, which lies in $[-1, 1]$: it is positive for favorable coverage, negative for unfavorable coverage, and near zero when the headline is neutral or the model is unsure. We keep a headline only when the model is confident, that is when its largest class probability is at least τ. The baseline keeps all scored headlines ($\tau = 0$) to maximize coverage, and we report sensitivity at $\tau = 0.6$ and $\tau = 0.9$.

The firm-day stance is the average of the headline scores for that firm on that day,

$$\text{stance}_{i,t} = \frac{1}{N_{i,t}} \sum_{a \in A_{i,t}} \left(p_a^+ - p_a^- \right), \tag{1}$$

where, for firm i on day t:

- $A_{i,t}$ is the set of retained headlines about firm i published on day t;
- $N_{i,t} = |A_{i,t}|$ is the number of such headlines, that is the daily coverage volume;
- $\text{stance}_{i,t} \in [-1, 1]$ is the average tone toward the firm on that day.

We keep the daily volume $N_{i,t}$ as a control in the regressions, because the average of a noisy score tightens mechanically as more headlines are averaged, and we want to separate a genuine change in tone from a change in the amount of coverage.

C. Market Alignment and Controls

The stance series is aligned to firm-day market outcomes: daily returns, the S&P 500 index return, and VIX, drawn from `stock_prices` and the VIX daily series. This alignment yields the panels used in the RQ1–RQ3 specifications.

D. Why a Regression Design

We test RQ1 and RQ2 with regression models rather than a raw comparison of pre- and post-2020 averages, because a raw comparison would mistake two persistent confounders for a genuine break. Firms differ permanently in how heavily the media covers them and in their average return, and on any given day all firms move together with the broad market. A regression that includes firm and time fixed effects removes both sources of confounding before we estimate the post-2020 change, and the next subsection states the assumptions under which that change is identified.

E. Identification and Panel Design

A regression with firm and time fixed effects nets out both confounders: it compares each firm against its own history while holding constant whatever is common across firms at each date, so the post-2020 coefficient measures the within-firm change net of these confounders. Identification rests on three assumptions. First, in the absence of the 2020 shock the treated and untreated periods would have followed a common trend, so the fixed effects capture the relevant heterogeneity. Second, the timing of the break is exogenous to firm-level coverage and returns, which the pandemic onset makes plausible. Third, after the within-transformation the residual shocks are uncorrelated with the post indicator; we report HC1-robust and, for RQ1, day-clustered standard errors to guard against the remaining heteroskedasticity and within-day dependence.

Both RQ1 and RQ2 use a pre/post design on a firm–day panel. The treatment indicator

$$\mathrm{Post}_t = \mathbf{1}[\, t \geq \text{2020-01-01}\,] \tag{2}$$

partitions the sample into a pre-period (2015–2019) and post-period (2020–2025). Firm fixed effects α_i absorb permanent cross-firm heterogeneity; time fixed effects δ_t absorb common shocks at the day or month level depending on the outcome. The stance outcome is the signed daily mean,

$$\mathrm{bias}_{it} = \frac{1}{|\mathcal{A}_{it}|} \sum_{a \in \mathcal{A}_{it}} (p_a^+ - p_a^-), \tag{3}$$

where $\mathcal{A}_{it}$ is the set of retained headlines about firm i on day t and p_a^+, p_a^- are the positive and negative stance probabilities for headline a defined in Section III-B. The returns outcome is the daily log return $\mathrm{return}_{it} = \ln P_{it} - \ln P_{i,t-1}$, where P_{it} is firm i's closing price on day t. Both equations are estimated by within-transformation OLS with HC1-robust standard errors; day-clustered standard errors are additionally reported for RQ1.

F. RQ1: Stance Regression

The dependent variable is the daily firm-level stance bias_{it}. We regress it on the post-2020 indicator together with firm and daily time fixed effects and a coverage-volume control:

$$\mathrm{bias}_{it} = \alpha_i + \delta_t^{\mathrm{day}} + \beta\, \mathrm{Post}_t + \gamma\, \mathrm{vol}_{it} + \varepsilon_{it} \tag{4}$$

Firm effects α_i absorb each firm's structural coverage level; daily time effects δ_t^{day} absorb market-wide sentiment common to all firms on a given day. The volume control $\mathrm{vol}_{it} = |\mathcal{A}_{it}|$ counts how many headlines firm i received on day t. We include it because averaging more headlines mechanically pulls the daily mean toward zero, so holding the count fixed separates a genuine change in tone from a change in the amount of coverage. We test

$$H_0^{(1)} : \beta = 0$$
$$H_1^{(1)} : \beta \neq 0$$

where β is the coefficient on Post_t.

G. RQ2: Returns Regression

The dependent variable is firm i's daily log return return_{it}. We regress it on the post-2020 indicator together with firm and monthly time fixed effects and two market controls:

$$\mathrm{return}_{it} = \alpha_i + \delta_t^{\mathrm{month}} + \beta\, \mathrm{Post}_t + \gamma_1\, \mathrm{sp500\_ret}_t + \gamma_2\, \mathrm{vix}_t + \varepsilon_{it} \tag{5}$$

Monthly time effects $\delta_t^{\mathrm{month}}$ absorb broad market-wide shifts that affect all firms in a given month, and the S&P 500 return and the VIX control for daily market movements and volatility within each month. We test

$$H_0^{(2)} : \beta = 0$$
$$H_1^{(2)} : \beta \neq 0$$

where β is the coefficient on Post_t.

Table III lists every variable that enters the RQ1 and RQ2 regressions, so the returns model in particular is not driven by stance alone but by the post-2020 indicator and the market controls together.

H. RQ3: VAR and Granger Causality

For the dynamic analysis we estimate a reduced-form bivariate VAR of order p on the endogenous vector $\mathbf{y}_t = (\Delta\mathrm{bias}_t,\, r_t)^\top$:

$$\mathbf{y}_t = \mathbf{c} + \sum_{\ell=1}^{p} \mathbf{A}_\ell\, \mathbf{y}_{t-\ell} + \mathbf{u}_t \tag{6}$$

where $\Delta\mathrm{bias}_t$ is the first-differenced weekly bias index (required for stationarity; see Section V) and r_t is the weekly log return. Writing $a_\ell^{(2,1)}$ for the coefficient on lag-ℓ stance in the return equation, we test

$$H_0^{(3)} : a_1^{(2,1)} = \cdots = a_p^{(2,1)} = 0$$
$$H_1^{(3)} : \exists\, \ell \text{ s.t. } a_\ell^{(2,1)} \neq 0$$

so that $H_0^{(3)}$ states that past stance does not help predict returns. We test it with a joint Wald F-test using Newey–West HAC standard errors.

IV. Results

A. RQ1: No Significant Aggregate Stance Shift

Table IV(a) reports estimates from Eq. (4) on $n = 27{,}627$ firm–day observations (26 firms, 3,982 days) built from the relevant headlines only. We fail to reject $H_0^{(1)}$: the post-shock coefficient is $\hat{\beta} = +0.0110$ with HC1 SE = 0.0296 ($p = 0.71$); day-clustered inference is nearly identical (SE = 0.0302, $p = 0.72$), confirming that within-day cross-firm dependence is not driving the result. Article volume is strongly negative (-0.0038^{***}): heavier-coverage firm-days regress toward neutral tone, validating the volume control. The within $R^2 = 0.0017$ reflects the well-known dominance of idiosyncratic headline noise at the firm-day level.

The null result is informative. A joint F-test of the 3,981 daily effects is jointly significant ($F = 1.05$, $p = 0.019$), meaning common daily shocks still account for a meaningful

TABLE III
VARIABLES IN THE RQ1 (STANCE) AND RQ2 (RETURNS) REGRESSIONS.

Variable	Used in	Description
bias_{it}	RQ1 (dep.)	Daily firm-level stance: signed mean of headline stance scores for firm i on day t.
return_{it}	RQ2 (dep.)	Daily log return of firm i, $\ln P_{it} - \ln P_{i,t-1}$.
Post_t	RQ1, RQ2	Treatment indicator, $\mathbf{1}[t \geq \text{2020-01-01}]$; its coefficient β is the parameter of interest.
vol_{it}	RQ1	Coverage volume: number of retained headlines about firm i on day t.
sp500_ret_t	RQ2	Daily S&P 500 index return, controlling for broad market movement.
vix_t	RQ2	Daily VIX level, controlling for market volatility.
α_i	RQ1, RQ2	Firm fixed effects, absorbing permanent cross-firm differences.
δ_t^{day}	RQ1	Daily time fixed effects, absorbing shocks common to all firms on a day.
δ_t^{month}	RQ2	Monthly time fixed effects, absorbing shocks common to all firms in a month.

Note: "(dep.)" marks the dependent variable; all other rows are regressors or fixed effects.

share of the time variation in stance. Once those shocks are absorbed by δ_t^{day}, the static Post step carries almost no residual identifying variation. We therefore fail to detect a systematic within-firm stance shift beyond what the broader news environment already explains.

B. RQ2: No Significant Shift in Firm-Specific Returns

Table IV(b) reports estimates from Eq. (5) on $n = 62{,}667$ firm–day observations (26 firms; pre-period mean returns tightly clustered, range 0.0019, SD 0.0004). We fail to reject $H_0^{(2)}$: once monthly time effects absorb common market-regime variation, the post-shock coefficient is $\hat{\beta} = -0.0018$ (SE $= 0.0142$, $p = 0.90$). The S&P 500 return regressor is absorbed by the month fixed effects ($\hat{\gamma}_1 \approx 0$, $p = 0.90$), and VIX is similarly uninformative within month ($p = 0.93$). The within $R^2 \approx 0$ follows directly: with the dominant systematic factor (index returns) loaded entirely onto the monthly FE, the regressors have negligible residual explanatory power.

Together, RQ1 and RQ2 establish a clean null baseline: after properly absorbing common daily and monthly shocks, neither media stance nor firm-level returns show a detectable aggregate level shift around January 2020. This motivates the dynamic question in Section V—whether the *lead–lag relationship* between stance and returns changed post-shock, even if the marginal distributions did not.

V. VECTOR AUTOREGRESSION AND GRANGER CAUSALITY ANALYSIS

The regressions in the previous section ask whether the level of each series shifts after 2020. They are silent on the dynamic relationship between media stance and returns: whether past stance helps predict future returns, whether returns feed back into coverage, and whether that lead–lag structure itself changed after the shock. We answer these questions with a vector autoregression (VAR) and Granger-causality framework, supported by an eight-stage data pipeline that constructs a clean, stationarity-validated weekly panel before any dynamic model is fit. We move to a weekly frequency for this analysis because daily firm-level stance is sparse and noisy for several firms, whereas weekly aggregation yields adequate coverage (median 92.9% of weeks across firms) while retaining enough observations (≈ 522 aligned weeks per firm) for lag-rich models. Per-firm results for every pipeline stage are tabulated in the Appendix.

A. The VAR Model

For a firm with endogenous vector $\mathbf{y}_t = (\Delta\text{bias}_t,\, r_t)^\top$, where Δbias_t is the first difference of the weekly bias index and r_t is the weekly log return, a reduced-form VAR of order p is

$$\mathbf{y}_t = \mathbf{c} + \sum_{\ell=1}^{p} \mathbf{A}_\ell\, \mathbf{y}_{t-\ell} + \mathbf{u}_t, \tag{7}$$

with coefficient matrices $\mathbf{A}_\ell \in \mathbb{R}^{2\times 2}$ and innovations $\mathbf{u}_t$ having covariance Σ. The off-diagonal entries of the $\mathbf{A}_\ell$ carry the cross-dynamics: a nonzero $(1,2)$ entry means past returns move current stance, and a nonzero $(2,1)$ entry means past stance moves current returns. The VAR is the minimal model that treats both variables as endogenous and lets the data, rather than an a priori causal ordering, reveal the direction of predictability, which is precisely what a media–market feedback question requires.

B. Stage 1–4: Building the Weekly Panel

The pipeline begins with a *data inventory* (Stage 1) that audits, for every candidate firm, article counts, coverage, and the share of zero-article days, and flags sparse firms (after relevance filtering the median firm posts an article on fewer than half of all trading days) for exclusion. This audit motivates the move to a weekly frequency and the retention of the sixteen firms that clear the coverage threshold. Stage 2 constructs the *weekly bias index* by aggregating signed article stances into ISO weeks and verifies distributional quality

TABLE IV
REGRESSION RESULTS FOR RQ1 (STANCE SHIFT, EQ. 4) AND RQ2 (RETURNS SHIFT, EQ. 5). BOTH SPECIFICATIONS FAIL TO REJECT THE NULL AT CONVENTIONAL LEVELS ONCE COMMON DAILY (RQ1) AND MONTHLY (RQ2) SHOCKS ARE ABSORBED.

(a) RQ1 — Daily Stance

Firm + day FE, HC1 SE

Variable	Coef.	SE	p
Post ($\hat{\beta}$)	+0.01096	0.02964	0.712
Article volume	−0.00381	0.00049	< 0.001
$n = 27{,}627$; 26 firms; 3,982 days			
Within $R^2 = 0.0017$			

(b) RQ2 — Daily Returns

Firm + month FE, HC1 SE

Variable	Coef.	SE	p
Post ($\hat{\beta}$)	−0.00184	0.01424	0.897
S&P 500 return	≈ 0	—	0.902
VIX	+0.00061	0.00654	0.925
$n = 62{,}667$; 26 firms			
Within $R^2 \approx 0$			

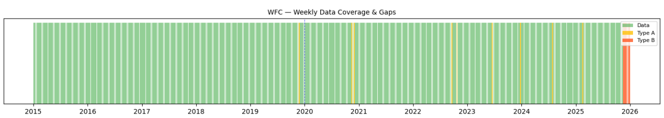

Fig. 4. Weekly data-coverage and gap map for Wells Fargo (WFC), the lowest-coverage retained firm. Short and medium gaps are imputed; the series still clears the $\geq 60\%$ coverage threshold.

(mean, skew, kurtosis, and the incidence of single-article weeks). Stage 3 performs *gap analysis and imputation*: weekly gaps are classified by length, with short Type-A gaps (1–2 weeks) forward-filled and medium Type-B gaps (3–8 weeks) linearly interpolated, while long Type-C gaps are left missing; 1,131 weekly observations are imputed (822 forward-fill, 309 interpolation) and all sixteen firms clear a $\geq 60\%$ coverage threshold with no gap exceeding eight weeks (the sparsest, Airbnb and Wells Fargo, near 69–71%). Crucially, both an imputed and an observed-only series are retained so that every downstream test can be re-run to confirm results are not artifacts of imputation. Figure 4 illustrates the coverage map for the sparsest retained firm.

Stage 4 constructs the *returns series* as weekly log returns from Monday-anchored ISO weeks and aligns them to the bias index. Log returns are used because they are additive across time and approximately symmetric, which suits a linear dynamic model. Quality checks flag extreme weekly moves ($> \pm 15\%$, concentrated in the March 2020 crash window) for later winsorization, and every firm clears the minimum of 100 aligned weeks, yielding a merged panel of 8,090 firm–week rows.

C. Stage 5: Stationarity and Cointegration

A VAR in levels is only valid if the series are stationary or jointly cointegrated; otherwise the regression is spurious. Each series is classified by combining the Augmented Dickey–Fuller (ADF) test, whose null is a unit root, with the KPSS test, whose null is stationarity. A series is called $I(0)$ when ADF rejects ($p < 0.05$) and KPSS does not, $I(1)$ in the opposite case, and "conflicting" (a likely structural break) when both reject. As expected for financial data, weekly log returns are $I(0)$ for essentially every firm, while on the

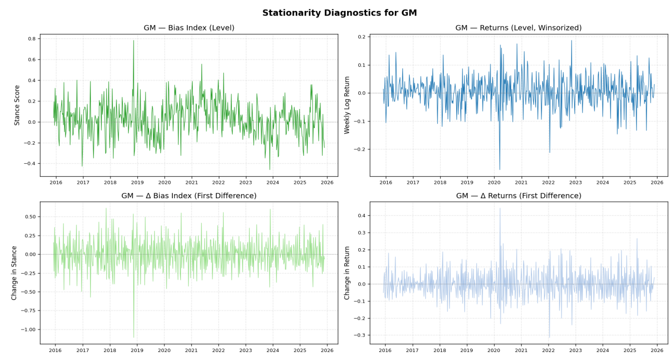

Fig. 5. Stationarity diagnostics for General Motors (GM): the bias index and weekly return in levels (left) and first differences (right), with ADF and KPSS verdicts driving the integration-order classification.

relevance-filtered series the bias index is $I(0)$ in levels for about half the firms and $I(1)$ or break-driven for the rest. To treat all firms under one specification and avoid a spurious levels regression where bias is $I(1)$, the VAR is specified on $(\Delta\text{bias}_t, r_t)$ rather than on the levels: differencing the bias index renders it stationary everywhere. A Johansen trace test detects a cointegrating relation for two firms, Boeing (trace $= 367.2$) and Wells Fargo (trace $= 317.5$), both far above the 95% critical value of 15.5, which would otherwise call for a VECM; these are flagged as robustness exceptions to the differenced specification. Figure 5 shows the level-versus-difference diagnostic for a representative firm.

D. Stage 6: Structural Break Testing

Because the central hypothesis concerns a regime change around 2020, we date breaks empirically rather than imposing the calendar. A Bai–Perron procedure, implemented by dynamic programming for a single change point under an L_2 (mean-shift) cost, locates the largest structural shift in the mean of each weekly series. The estimated bias-index break dates serve as the `is_post_break` cut used to split the sample for the subsequent subsample Granger tests, and every firm yields a valid split with at least 60 observations on each side. Figures 6 and 7 show the data-driven breaks for two firms; several cluster near the COVID window while others

Fig. 6. Bai–Perron structural breaks for Visa (V). The return break (2020-03-23) coincides with the COVID crash; the bias-mean break (2024-06-10) is firm-specific and later.

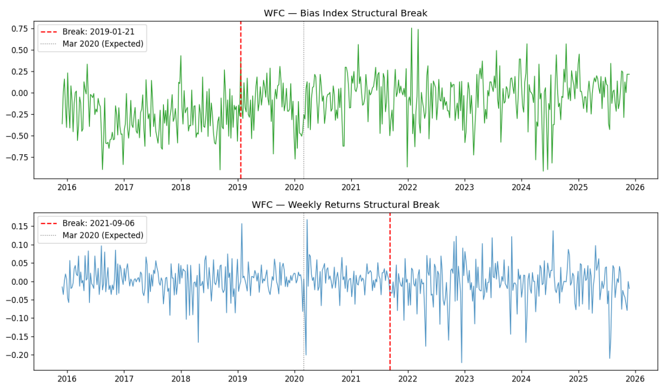

Fig. 7. Bai–Perron structural breaks for Wells Fargo (WFC), one of two firms with a cointegrating bias–return relation; its return-to-bias feedback emerges only post-break.

reflect firm-specific events, which is itself evidence that a single imposed 2020 dummy would be too blunt.

E. Stage 7: VAR Specification

For each firm the VAR order is selected by estimating Eq. (6) for $p = 1, \dots, 8$ and minimizing the Bayesian Information Criterion (BIC), which penalizes parameters more heavily than AIC and so favors parsimonious, well-identified models; when BIC selects zero lags we impose $p = 1$ to retain a dynamic model. The selected orders range from 1 to 6 (modal value $p = 3$). Three residual diagnostics certify each fit. Stability requires all roots of the companion matrix to lie outside the unit circle, equivalent to $\det(\mathbf{I} - \sum_\ell \mathbf{A}_\ell z^\ell) \neq 0$ for $|z| \leq 1$; every selected model is stable (minimum companion root > 1.19). The Portmanteau test checks for residual autocorrelation, confirming the chosen lag is adequate for most firms. The Jarque–Bera test rejects normality everywhere, the familiar heavy-tail signature of financial data, which leaves VAR coefficients consistent but motivates robust inference; Engle's ARCH test detects volatility clustering in several return residuals, reinforcing the use of HAC-robust standard errors in the causality stage.

TABLE V
FIRM-LEVEL GRANGER CAUSALITY, FULL SAMPLE (HAC p-VALUES)

Firm	Lag	Bias→Ret	Ret→Bias
UBER	6	0.003	0.375
BA	5	0.097	0.119
GS	3	0.432	0.004

Other 12 firms: no direction significant at 5%.

F. Stage 8: Granger Causality

Granger causality operationalizes "does past X help predict Y beyond Y's own past." Stance Granger-causes returns if the lagged stance coefficients are jointly nonzero in the return equation, i.e. $H_0 : a_1^{(2,1)} = \cdots = a_p^{(2,1)} = 0$ is rejected. We test this with a direct OLS regression of each variable on p lags of both variables, using a joint Wald (F) test and Newey–West HAC-robust standard errors (`maxlags`$= p$) to neutralize the heteroscedasticity and autocorrelation found in Stage 7. Table V reports the firm-level full-sample p-values. Causality is sparse in the full sample, as expected if the relationship is regime-dependent: stance Granger-causes returns for Uber ($p = 0.003$) and weakly for Boeing ($p = 0.097$), while the reverse direction (returns to bias) is significant for Goldman Sachs ($p = 0.004$).

The more informative test splits each firm at its empirical bias break and re-estimates causality on each side. Several firms that are quiet in the full sample show post-break causality that is absent pre-break: Uber is the clearest, with stance Granger-causing returns at $p < 0.001$ post-break versus $p = 0.48$ pre-break, and Intel marginally so ($p = 0.074$ post). Feedback (returns→bias) emerges post-break for AT&T ($p = 0.015$), Boeing ($p = 0.010$), and Wells Fargo ($p = 0.030$), while Goldman Sachs and Morgan Stanley show it only pre-break. This pre/post asymmetry is the firm-level analogue of the structural shift the regressions probe: the media–market link is not constant but switches on after the break for a subset of firms.

G. Panel VAR

Firm-by-firm tests are underpowered because each uses only one short series. We therefore pool the cross-section into a panel VAR, which is the design the study most relies on for its dynamic conclusion. Unobserved firm heterogeneity is removed with the Arellano–Bover/Helmert transformation (forward mean-differencing), which subtracts from each observation the mean of all its *future* values within the firm,

$$\tilde{y}_{it} = c_{it}\Big(y_{it} - \tfrac{1}{T_{it}} \textstyle\sum_{s>t} y_{is}\Big), \tag{8}$$

with a scaling factor c_{it} that equalizes the variance of the transformed errors. Unlike ordinary within-demeaning, this transformation preserves the orthogonality between the transformed variables and the lagged regressors, avoiding the look-ahead bias (Nickell bias) that contaminates dynamic fixed-effects panels. The endogenous variables are again Δbias and the weekly log return; we additionally partial out the exogenous

TABLE VI
PANEL VAR GRANGER CAUSALITY (15 FIRMS, HELMERT-TRANSFORMED, HAC)

Sample	N	Bias→Ret	Ret→Bias
Full	7493	0.565	0.899
Pre-break	4415	0.301	0.861
Post-break	3018	0.905	0.937

controls Post_t, VIX, the S&P 500 weekly log return, and the article count N_{articles}, so that any detected causality cannot be attributed to market-wide volatility or coverage volume. With $p = 3$ lags and HAC-robust standard errors, the pooled model is estimated by OLS on the transformed, constant-free data, and Granger directions are tested with joint Wald statistics over the full, pre-break, and post-break samples. Table VI reports the results.

Pooling buys power but, even so, neither Granger direction is significant at conventional levels in any subsample once VIX, the index return, and article volume are controlled. The honest reading is that the firm-level post-break signals (notably Uber, with scattered feedback effects elsewhere) do not aggregate into a uniform panel-wide media-to-returns channel: the dynamic link, where it exists, is firm-specific and heterogeneous rather than a market-wide regularity. This strengthens rather than weakens the contribution, because the panel VAR is exactly the test that would manufacture a spurious aggregate effect if the controls were omitted, and it declines to do so.

H. Limitations of the VAR Pipeline

Several caveats bound the dynamic findings. A weekly frequency, chosen for coverage, may be too coarse to capture the intraday or daily horizon over which news is plausibly priced, so a true fast media-to-price channel could be aliased away by aggregation. The single-break Bai–Perron design dates exactly one change point per series; firms with multiple regime shifts are only partially described, and the break date is estimated with sampling error that the subsample split treats as known. The bivariate VAR omits other drivers of both stance and returns (earnings surprises, analyst actions, sector shocks), so the controls in the panel model mitigate but do not eliminate omitted-variable concerns. Heavy-tailed, volatility-clustered residuals mean small-sample inference leans on the HAC correction and asymptotics rather than exact distributions. Finally, Granger causality is predictive, not structural: a significant test says past stance forecasts returns, not that stance mechanically moves prices, and the firm-specific nature of the surviving signals counsels caution against a universal causal claim.

VI. IDENTIFICATION, VALIDITY, AND ROBUSTNESS

Three threats bear on the level-shift findings. First, macro shocks move both coverage and prices; the day and month fixed effects absorb common shocks, and the S&P 500 return and VIX controls strip the systematic component out of returns. Second, article volume differs widely across firms (Fig. 3); we hold it fixed with the volume covariate, and substituting $\ln(1 + \text{vol})$ for raw volume leaves the RQ1 estimates essentially unchanged. Third, the pre/post design assumes within-firm parallel trends; the flat pre-shock coefficients in the RQ1 monthly event study and the tightly clustered pre-period return means (range 0.0019) are consistent with this premise, though they do not prove it. We further confirm the RQ1 conclusion is robust to the stance confidence threshold ($\tau = 0, 0.6, 0.9$) and certify the estimator with the standard battery (Breusch–Pagan, Jarque–Bera, VIF, Durbin–Watson), which motivates the HC1 and day-clustered inference reported throughout. Two limitations remain: the stance signal inherits measurement noise from the upstream NewsMTSC and relevance stages, which attenuates coefficients toward zero, and a binary post indicator cannot separate the 2020 shock from other contemporaneous regime changes.

VII. CONCLUSION

We assembled a target-dependent media-stance signal for 26 S&P 500 firms from 90,579 relevance-filtered headlines and tested whether the 2020 shock moved stance, returns, and the link between them. The level-shift results are null: once common daily and monthly shocks are absorbed by two-way fixed effects, neither media stance (RQ1, $\beta = +0.011$, $p = 0.71$) nor firm-specific returns (RQ2, $\beta = -0.002$, $p = 0.90$) show a significant aggregate shift around January 2020. The dynamic analysis (RQ3) reinforces this reading: a weekly panel VAR finds no market-wide Granger link between stance and returns in any subsample once volatility, the index return, and coverage volume are controlled, with the surviving predictability confined to a few firms (notably Uber, with scattered feedback elsewhere) rather than the panel as a whole. Taken together, the evidence points to a media–market relationship that is firm-specific and heterogeneous rather than a uniform post-2020 regularity—a conclusion that the controlled panel design supports precisely because it declines to manufacture a spurious aggregate effect.

APPENDIX

Table VII reports the decision-threshold sweep for the chosen single model on the held-out test set, which underlies Figure 2. Precision and recall trade off monotonically with the threshold, and a value near 0.80 is the operating point that satisfies both targets.

Table VIII reports per-firm activity for the sixteen firms carried into the weekly modeling panel, on the relevance-filtered corpus. After relevance filtering the median firm posts well under one relevant article per trading day, which is what drives the move to a weekly frequency; at the weekly level coverage is adequate (median 92.9% of weeks) though several firms (Airbnb, Bank of America, Wells Fargo) sit near 70%. Sparse names excluded upstream (e.g. MMM, AAPL, ETN, NDAQ, NKE, ORCL, PGR, TSLA, FDX) are omitted.

Weekly gaps are classified by length: Type A (1–2 weeks) are forward-filled and Type B (3–8 weeks) linearly inter-

TABLE VII
HELD-OUT THRESHOLD SWEEP FOR THE CHOSEN SINGLE MODEL ($n = 190$). THE ADOPTED OPERATING POINT IS SHOWN IN BOLD.

Threshold	P	R	F1
0.50	0.878	0.857	0.867
0.70	0.875	0.833	0.854
0.75	0.897	0.833	0.864
0.80	**0.919**	**0.810**	**0.861**
0.85	0.914	0.762	0.831
0.90	0.912	0.738	0.816
0.92	0.968	0.714	0.822
0.95	1.000	0.714	0.833

TABLE VIII
PER-FIRM COVERAGE IN THE MODELING PANEL (RELEVANCE-FILTERED)

Ticker	Avg. Art./Day	Zero Days (%)	Weekly Cov. (%)
ABNB	0.472	78.7	68.6
AMZN	1.966	42.6	97.4
T	3.340	29.1	98.8
BA	1.334	53.7	95.3
BAC	0.357	81.7	71.3
GM	2.012	50.0	96.2
GS	0.866	60.5	94.6
INTC	1.205	58.6	92.9
MCD	0.705	73.1	79.8
MSFT	2.817	34.1	99.0
MS	0.813	60.0	95.5
SBUX	0.705	75.8	73.0
UBER	2.031	56.2	89.2
V	0.996	56.5	94.1
WMT	0.977	64.9	87.8
WFC	0.875	75.8	70.9

polated, while Type C ($>$ 8 weeks) are left missing. On the relevance-filtered series 1,131 weekly observations are imputed (822 forward-fill, 309 interpolation); every firm still clears the $\geq 60\%$ coverage threshold with no gap exceeding eight weeks (Table IX). The coverage map for the sparsest retained firm is shown in Fig. 4.

Weekly log returns are computed from Monday-anchored ISO weeks and aligned to the bias index (Table X). Extreme weekly moves ($> \pm 15\%$) cluster in the March 2020 crash window. Returns are winsorized at ± 5 standard deviations from each firm's mean, with AT&T (T) excluded from winsorization so that its outlier behavior can be compared against the others.

Integration orders combine the ADF and KPSS verdicts on the levels and first differences of each series (Table XI). Returns are $I(0)$ almost everywhere; on the relevance-filtered series the bias index is now $I(0)$ in levels for about half the firms and $I(1)$ or break-driven for the rest. A Johansen trace test detects cointegration for two firms, Boeing (trace $=$ 367.23) and Wells Fargo (trace $=$ 317.53), both far above the 95% critical value of 15.49; these are the VECM candidates noted as robustness exceptions to the uniformly differenced specification.

Lag orders are chosen by BIC over $p = 1, \ldots, 8$; Table XII

TABLE IX
WEEKLY GAP CLASSIFICATION AND IMPUTATION COUNTS

Ticker	Cov. (%)	Gaps A	Gaps B	Longest	Imputed
ABNB	68.6	84	18	7	180
AMZN	97.4	11	1	4	15
T	98.6	1	1	7	8
BA	95.3	27	0	1	27
BAC	71.1	104	10	5	166
GM	96.0	20	0	2	23
GS	94.6	29	0	2	31
INTC	92.9	34	2	3	41
MCD	79.8	82	4	6	116
MSFT	98.8	7	0	1	7
MS	95.3	27	0	1	27
SBUX	73.0	89	12	7	155
UBER	89.0	41	2	8	63
V	93.9	25	1	7	35
WMT	87.8	52	2	8	70
WFC	70.9	69	20	8	167

reports the AIC and BIC choices, the selected order, and residual diagnostics. Every selected model is stable (minimum companion root > 1). Jarque–Bera rejects normality everywhere (heavy tails), and Engle's ARCH test flags volatility clustering in most return residuals, together motivating the HAC-robust causality tests. WMT is excluded from this specification pass.

Bai–Perron single-break dates for each weekly series, with the resulting pre/post observation counts, are in Table XIII. The bias-index break defines the subsample split for the causality tests. General Motors breaks at 2020-01-13, closest to the COVID shock; most other bias breaks are firm-specific.

Table XIV reports HAC Granger p-values for every firm, in both directions, over the full sample and the pre/post subsamples defined by the bias break. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table XV summarizes the classical-linear-model diagnostics for the RQ1 two-way fixed-effects estimator. Orthogonality and conditioning are satisfied; heteroscedasticity and non-normality are present and handled by the HC1/clustered errors and large-sample asymptotics, respectively; collinearity is negligible.

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TABLE X
WEEKLY LOG-RETURN QUALITY AND WINSORIZATION (16 FIRMS)

Ticker	Weeks	Mean	Std	Min	Max	Extreme ($>\pm15\%$)	Winsorized
ABNB	522	0.00265	0.03562	−0.1443	0.1399	0	0
AMZN	522	0.00235	0.03230	−0.1829	0.1138	1	1
T	522	0.00147	0.06861	−0.4093	0.3331	18	skipped
BA	522	−0.00046	0.03924	−0.2849	0.1939	4	1
BAC	260	−0.00090	0.06274	−0.1815	0.1896	5	0
GM	522	0.00503	0.05559	−0.2923	0.1868	6	1
GS	522	0.00298	0.03935	−0.2026	0.1322	2	1
INTC	522	0.00224	0.03584	−0.1571	0.1397	1	0
MCD	522	0.00251	0.04381	−0.2056	0.2160	5	0
MSFT	522	0.00209	0.04499	−0.2197	0.1940	6	0
MS	522	0.00335	0.04431	−0.2955	0.2533	6	2
SBUX	522	0.00264	0.04417	−0.1883	0.1934	3	0
UBER	522	0.00250	0.04123	−0.2416	0.2012	4	1
V	522	0.00154	0.05526	−0.2859	0.1954	6	1
WMT	522	0.00154	0.05526	−0.2859	0.1954	6	1
WFC	522	0.00012	0.04667	−0.2215	0.1680	10	0

TABLE XI
INTEGRATION ORDER OF BIAS AND RETURN SERIES

Ticker	(Bias, Return)	Note
ABNB	$(I(0), I(0))$	both stationary
AMZN	$(I(1), I(0))$	
T	$(I(0), I(0))$	both stationary
BA	$(I(1), I(1))$	cointegrated (VECM)
BAC	$(I(1), I(0))$	
GM	$(I(0), I(0))$	both stationary
GS	$(I(1), I(0))$	
INTC	$(I(0), I(0))$	both stationary
MCD	$(I(0), I(0))$	both stationary
MSFT	$(I(1), I(0))$	
MS	$(I(0), I(0))$	both stationary
SBUX	$(I(1), I(0))$	
UBER	$(I(1), I(0))$	
V	$(I(0), I(0))$	both stationary
WMT	$(I(0), I(0))$	both stationary
WFC	$(I(1), I(1))$	cointegrated (VECM)

TABLE XII
VAR LAG SELECTION AND RESIDUAL DIAGNOSTICS (15 FIRMS)

Ticker	AIC	BIC	Chosen	N	Min Root	Portm. p	JB p	ARCH(B) p	ARCH(R) p
ABNB	5	3	3	521	1.62	0.0356	0.0000	0.0097	0.0679
AMZN	7	2	2	521	1.93	0.0002	0.0000	0.0202	0.0000
T	7	6	6	521	1.24	0.2131	0.0000	0.2725	0.7046
BA	8	5	5	521	1.29	0.2558	0.0000	0.0120	0.0000
BAC	5	1	1	259	3.09	0.2267	0.0000	0.8870	0.2809
GM	8	4	4	521	1.39	0.0416	0.0000	0.0662	0.0004
GS	7	3	3	521	1.48	0.0038	0.0000	0.7429	0.0003
INTC	8	3	3	521	1.58	0.0139	0.0000	0.0363	0.0000
MCD	8	3	3	521	1.56	0.0016	0.0000	0.5908	0.0000
MSFT	8	3	3	521	1.53	0.0114	0.0000	0.0134	0.2055
MS	8	4	4	521	1.42	0.0024	0.0000	0.0645	0.0000
SBUX	6	3	3	521	1.55	0.0475	0.0000	0.7787	0.0000
UBER	8	6	6	521	1.19	0.0042	0.0000	0.0001	0.0000
V	7	3	3	521	1.56	0.0033	0.0000	0.0032	0.0000
WFC	8	1	1	521	2.57	0.0003	0.0000	0.0410	0.0000

TABLE XIII
EMPIRICAL STRUCTURAL BREAK DATES

Ticker	Bias Break	Return Break	Pre	Post
ABNB	2023-02-27	2021-09-06	378	144
AMZN	2018-10-22	2022-01-03	151	371
T	2022-11-28	2021-09-27	365	157
BA	2019-03-04	2022-01-03	170	352
BAC	2022-05-02	2022-12-26	73	187
GM	2020-01-13	2017-10-30	215	307
GS	2024-01-15	2021-11-22	424	98
INTC	2024-07-22	2023-03-20	451	71
MCD	2020-06-29	2023-10-30	239	283
MSFT	2022-05-30	2021-09-27	339	183
MS	2017-07-03	2023-10-30	83	439
SBUX	2022-02-07	2022-10-03	323	199
UBER	2023-01-16	2024-04-01	372	150
V	2024-06-10	2020-03-23	445	77
WMT	2024-04-22	2020-03-23	438	84
WFC	2024-10-07	2021-09-06	462	60

TABLE XIV
FIRM-LEVEL GRANGER CAUSALITY, FULL AND SUBSAMPLE (HAC p-VALUES)

		Full		Pre-break		Post-break	
Ticker	Lag	B→R	R→B	B→R	R→B	B→R	R→B
ABNB	3	0.362	0.892	0.165	0.884	0.416	0.936
AMZN	2	0.944	0.277	0.228	0.795	0.772	0.137
T	6	0.570	0.367	0.525	0.670	0.484	0.015**
BA	5	0.097*	0.119	0.091*	0.247	0.258	0.010**
BAC	1	0.945	0.550	0.578	0.407	0.778	0.849
GM	4	0.629	0.543	0.805	0.515	0.498	0.519
GS	3	0.432	0.004***	0.397	0.026**	0.919	0.210
INTC	3	0.611	0.363	0.668	0.535	0.074*	0.157
MCD	3	0.276	0.518	0.178	0.456	0.920	0.461
MSFT	3	0.537	0.847	0.157	0.060*	0.981	0.721
MS	4	0.389	0.959	0.019**	0.514	0.462	0.938
SBUX	3	0.419	0.207	0.187	0.286	0.682	0.542
UBER	6	0.003***	0.375	0.478	0.270	0.000***	0.131
V	3	0.281	0.860	0.253	0.871	0.854	0.268
WFC	1	0.303	0.148	0.593	0.497	0.138	0.030**

TABLE XV
RQ1 SUMMARY DIAGNOSTICS

Test	Statistic	Implication
Residual mean	-9.0×10^{-19}	orthogonality OK
Condition number	157.46	numerically stable
Breusch–Pagan (LM)	190.25	heteroscedastic (HC1 used)
Jarque–Bera	4541.18	fat tails (n large)
VIF (both regressors)	≈ 1.00	no collinearity
Max pairwise corr.	0.0189	negligible
Durbin–Watson	1.673	mild autocorrelation